Analytical Cheistry 1 (125101)

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Framework of this course

❖ Introduction (L1)

❖ Section I Chemical Measurements

1.1 Basic knowledge (L2)

1.2 Error and estimation (L3&L4)

1.3 Linear regression (L5)

❖ Section II Chemical Equilibrium

2.1 pH and buffer solutions (L6)

2.2 Titrations (L7&L8)

2.3 Potentiometry in titrations (L9)

2.4 Gran plot in titrations (L10)

2.5 Electroanalysis (L11)

2.6 Gravimetric analysis (L12)

❖ Summary (L13)

2.6 Gravimetric analysis

2.6 Gravimetric analysis

Overview

2.6-1 Precipitation process

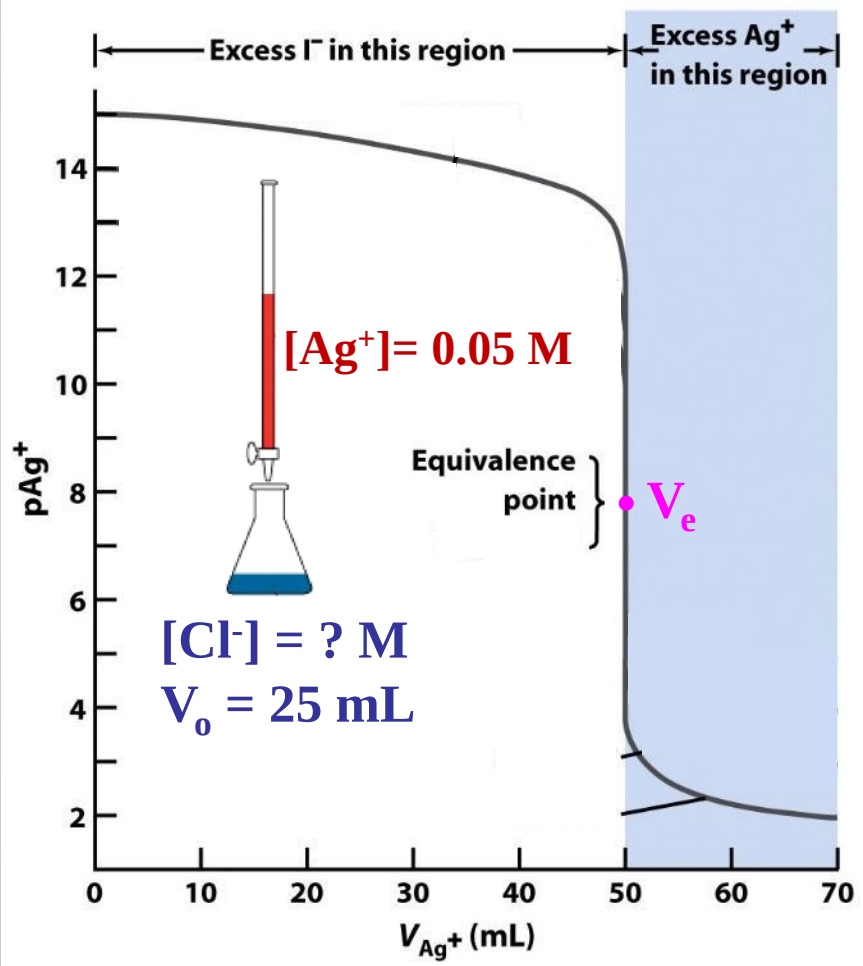
2.6-2 Homogeneous precipitation

2.6-3 Colloidal precipitates

2.6-4 Precipitate washing

Precipitation titration

Titration based on the reaction in which the product of precipitates



Find the unknown of $[Cl^-]$ from the V_e of the titrant of $[Ag^+]$

- Another methods?

Gravimetric analysis

Definition:

Quantitative analysis based on weighing a precipitate

Example: Analysis of chloride

$$Cl^- + Ag^+ \rightarrow AgCl\downarrow$$

Unknown quantity	Precipitant (in excess)	Precipitate (Weighing)

Gravimetric calculation relate moles of product to moles of analyte.

Procedure stages in gravimetric analysis

- Sampling
- Weighing the sample
- Sample dissolution
- Adding precipitant agent ($AgNO_3$) in excess
- Precipitating of AgCl
- Filtering the precipitate
- Washing the precipitate
- Drying the precipitate
- Weighing the precipitate
- Calculating the quantity of Cl^-
- Presentation of results

Procedure stages affecting the accuracy

1. Reagent concentration
2. Reagent mixing
3. Reaction temp.
4. Precipitation duration
5. Washing liquid used
6. Drying temp.

How do they influence accuracy?

The aim for precipitation

- Larger particles
- Purer particles
- Easy to filter

Understanding precipitation process

2.6-1 Precipitation process

Experimental stages in precipitation:

(a) Induction stage

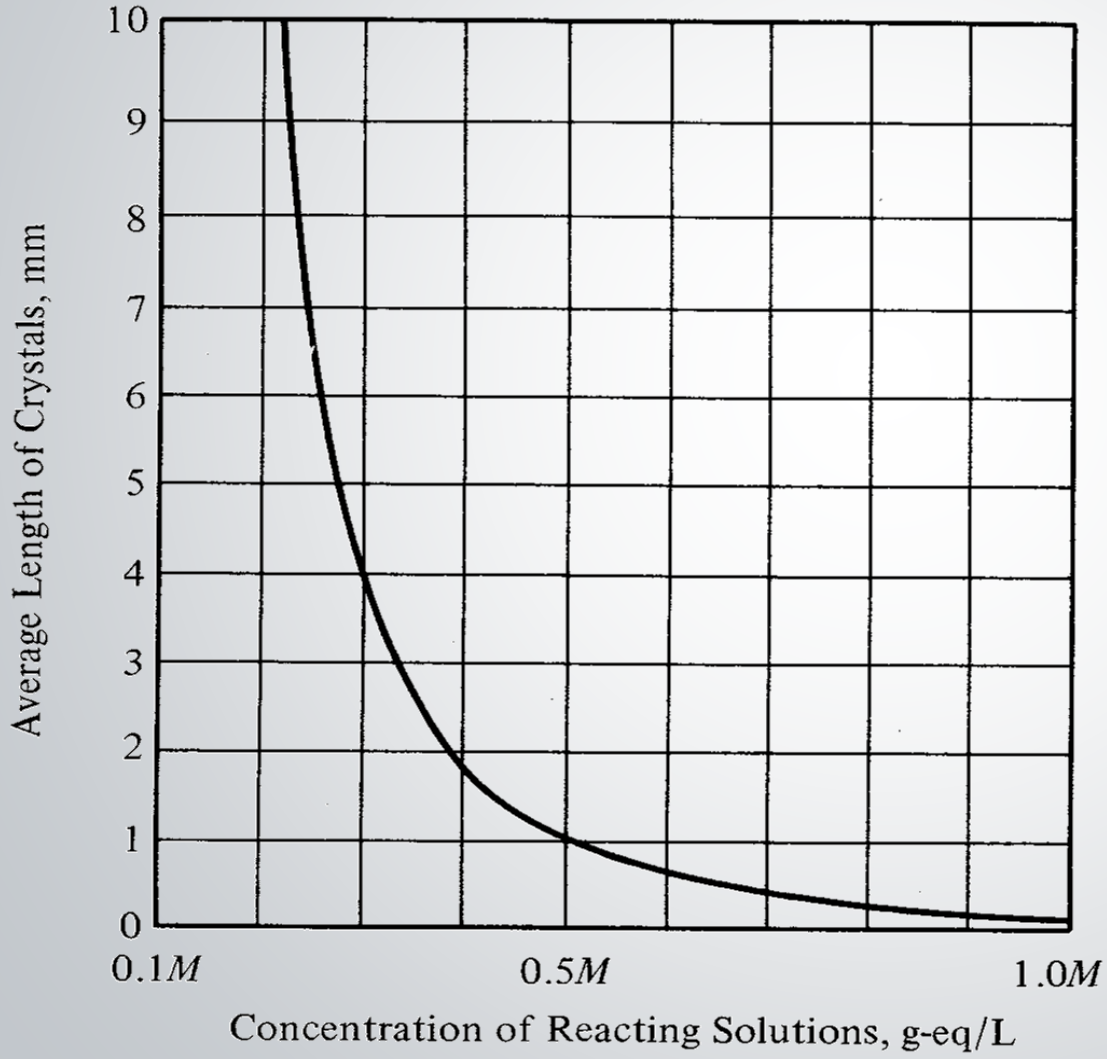
(Time from mixing reagents to nucleation starting)

(b) Nucleation stage

(c) Crystallite growth

What determines the final crystal size?

Crystal size vs. initial concentration of solution



Conclusions:

Concentrated solutions
⇒ small crystals.

Diluted solutions
⇒ large crystals.

Von-Weimarn found that the final crystal size is inversely proportional to the **relative supersaturation** of the solution

$$\textbf{Relative Supersaturation} = \frac{Q-S}{S}$$

Q : **initial** reagent concentration (just after mixing of reagents and before precipitation takes place)

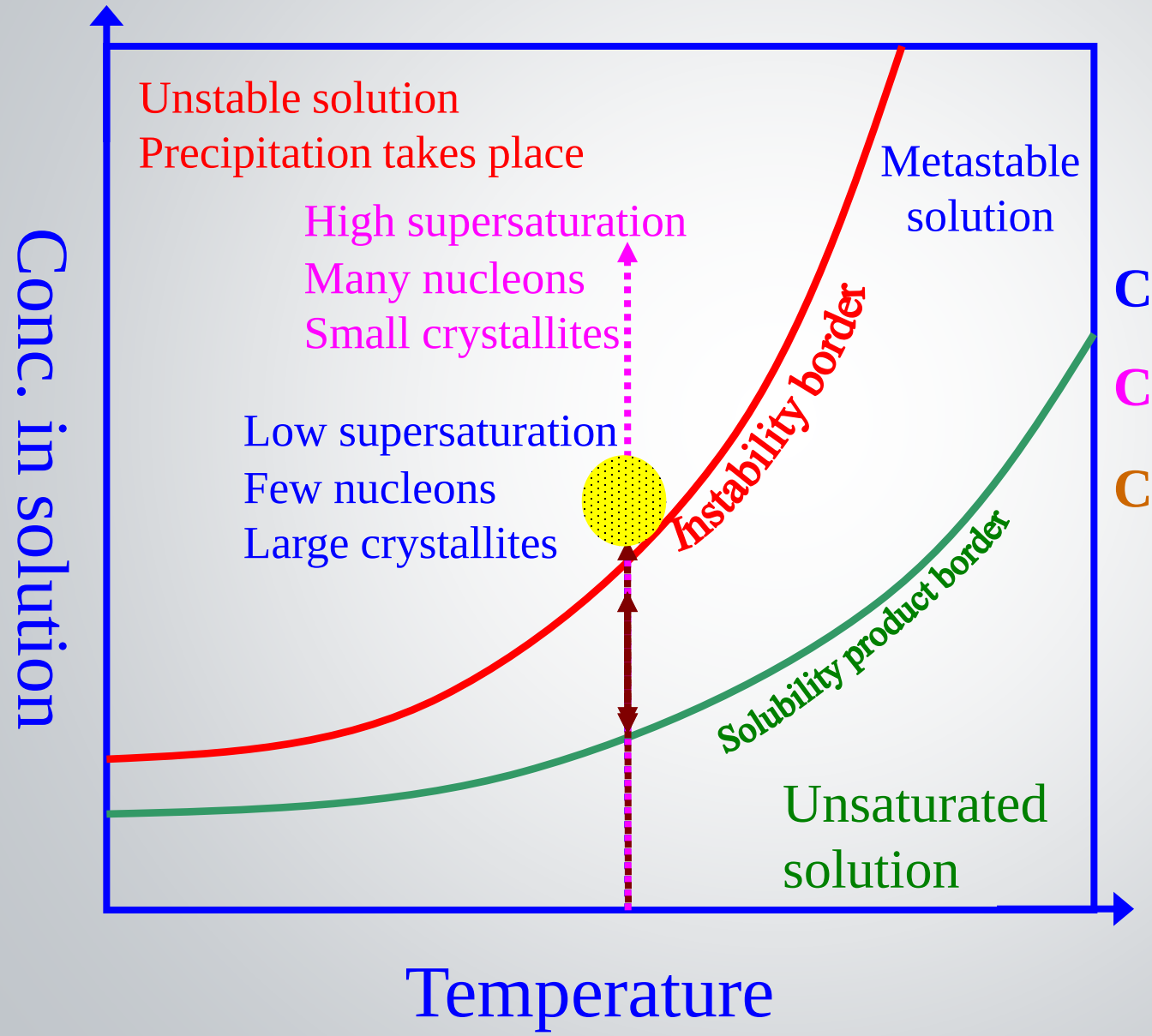
S : **final** solution concentration (at equilibrium conditions)

As the relative supersaturation is **lower**, the crystals are **larger**

Optimal precipitation for gravimetric analysis

- Slowly increase concentration to instability border. A few nucleons are formed.
- Add reagents during the crystallization, keeping the concentration within the metastable band. This way, no new nucleons are formed and the final crystals are large.

Optimal precipitation for gravimetric analysis



Case a: low Q-S

Case B: high Q-S

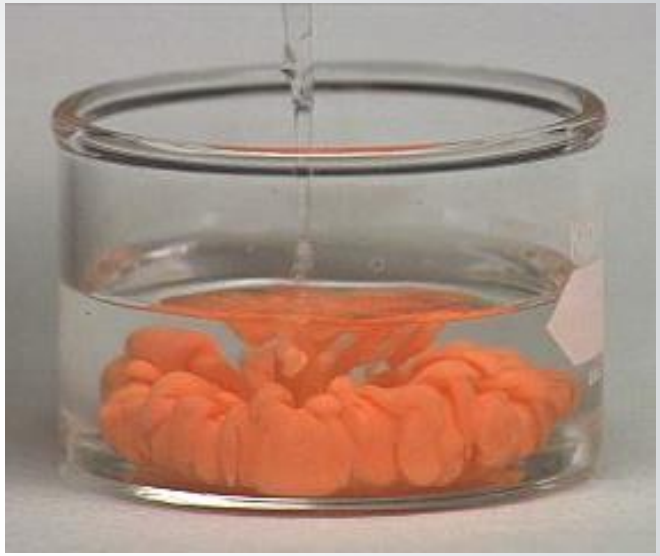
Case C: optimal

Problem: High local and temporal conc. during reagent mixing

High (Q-S) → many nucleons

$$K_2CrO_4 + Ba(NO_3)_2 \rightarrow BaCrO_4\downarrow$$

$$HgCl_2 + 2\ NaI \rightarrow HgI_2(s) + 2\ NaCl$$



Solution: Homogeneous precipitation

(Q-S) is kept low
to obtain large crystals

2.6-2 Homogeneous precipitation

1. If the precipitation is rapid, nucleation tends to dominate, resulting in many small particles
2. Make the precipitation slow, so that the particle growth dominates over the nucleation, and obtain larger particles

Main methods

- Precipitant is produced slowly from an initially homogeneous solution by chemical reaction.

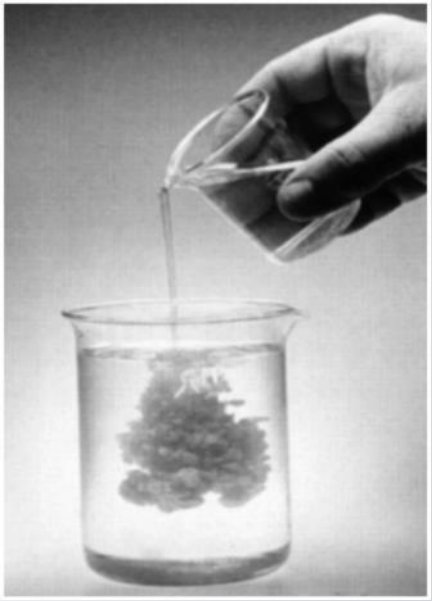
The difference between regular and homogeneous precipitations

Precipitation of $Fe(OH)_3$

2.6-3 Crystalline or colloidal precipitates?

- Crystalline precipitate (> 0.1 mm)
- Colloidal precipitate (1 – 1000 nm)

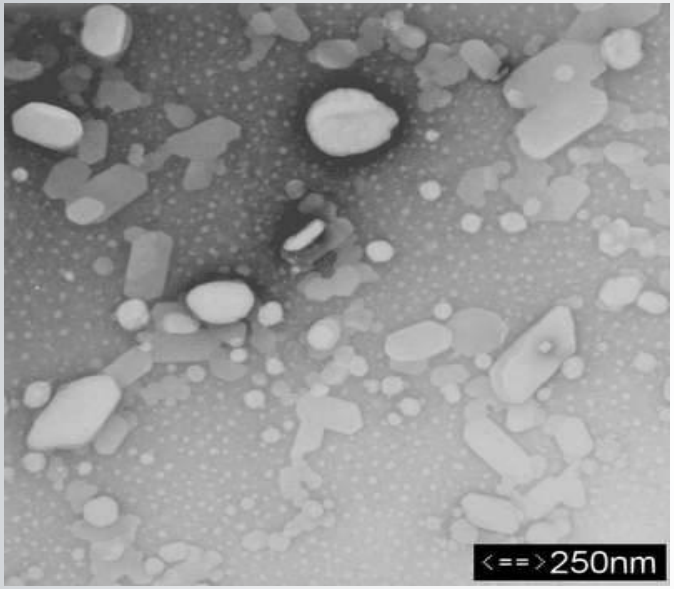
Crystalline



Colloidal

Colloids are larger than most molecules but too small to crystal.

How to determine a colloidal solution?



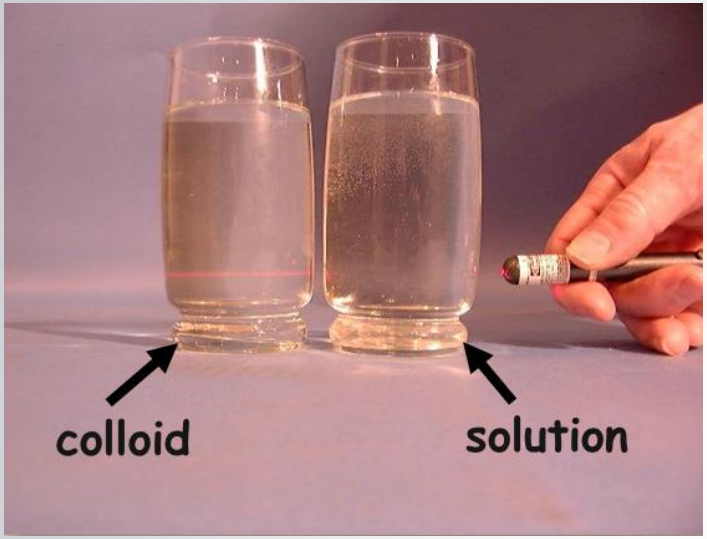
Colloidal precipitation

Tyndall effect

Colloids are suspended by Brownian motion (random movement) of solvent molecules.

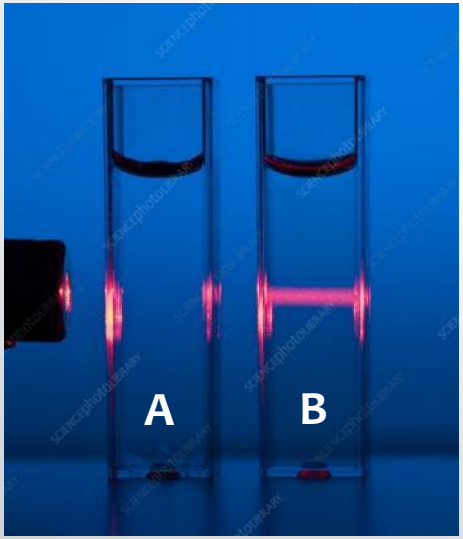
Tyndall effect is an easy way of determining a mixture is colloidal or not, based on the light scattering.

How to determine a colloidal solution?



Colloids scatter light, making a beam visible. Solutions do not scatter light.

Which glass contains a colloid?



Crepuscular (Jesus) rays

God's Eye; Jacob's Ladder

The colloidal dispersion is the dust or droplet (Aerosol)

Very low-solubility materials (low S) such as $Fe(OH)_3$ precipitate at high relative supersaturation and form colloids

What's the problem with colloids?

- Cannot be filtered (almost)
- Cannot be weighted (almost)

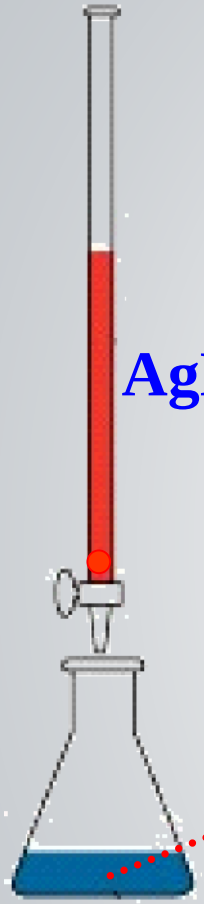
What's the solution?

Understanding the colloidal structure

⬇

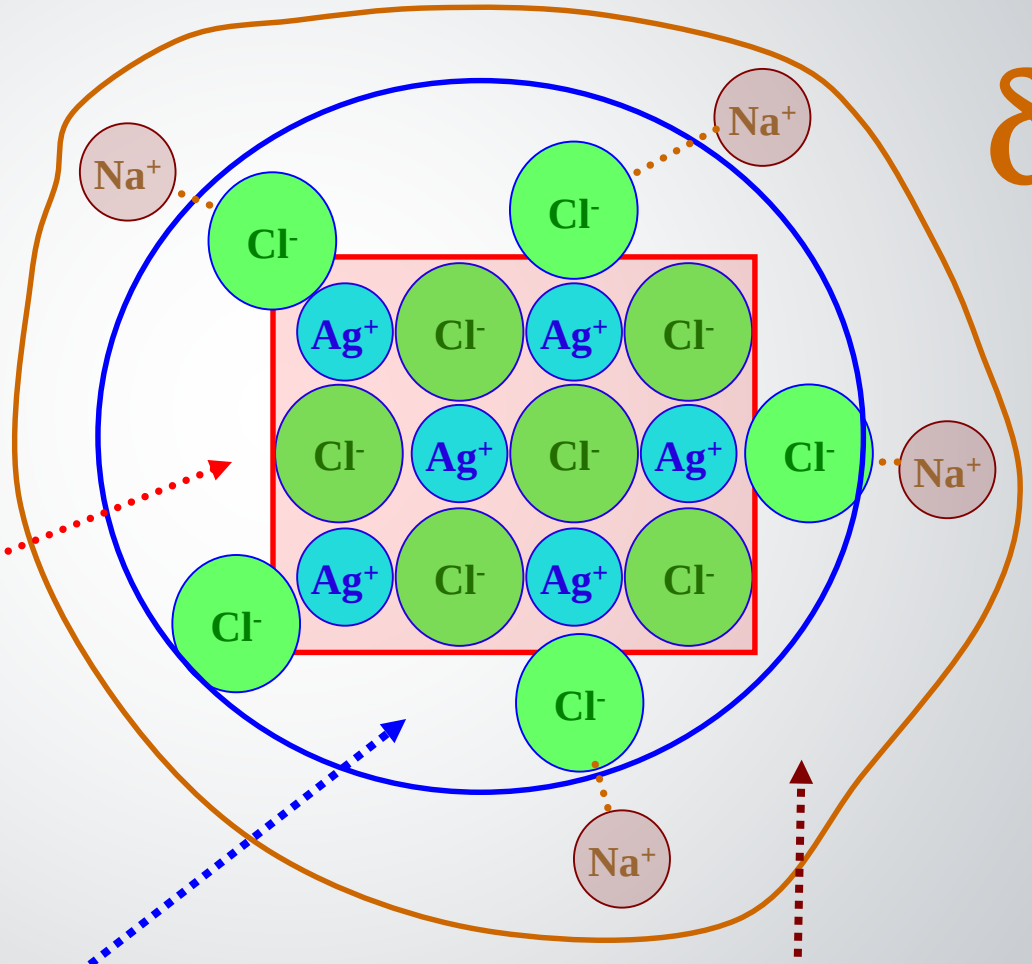
Optimal precipitation procedure

The electric double layers for colloid



$AgNO_3$

NaCl (in excess)



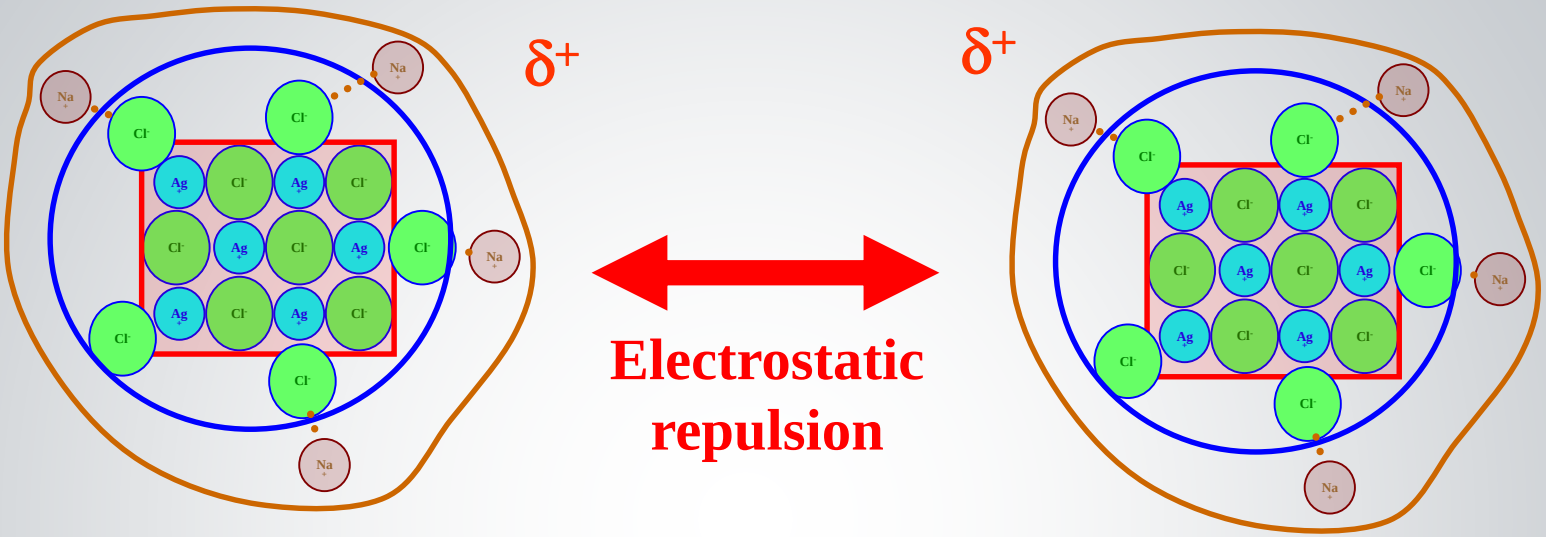
1st ionic layer of negatively charged particle

2nd ionic layer of positively charged ionic atmosphere

The electric double layers

2nd ionic layer	1st ionic layer
Charge opposite to the 1st layer	Crystal ion which is in excess
Weak electrostatic bonds to the 1st layer	Strongly bonded to crystal

Colloidal particles repel one another



Particles must have enough kinetic energy to overcome electrostatic repulsion to coalesce

- ✓ **Ending precipitation with no reagent excess**
- ✓ **Heating promotes coalescence through collision**
- ✓ **Increasing electrolyte concentration**

The dielectric constant (ξ) increase with the concentration of electrolyte, thus the electrostatic forces F will become less ($F = \kappa\frac{\delta_1\delta_2}{r^2}$; $\kappa = \frac{1}{4\pi\xi}$).

Examples

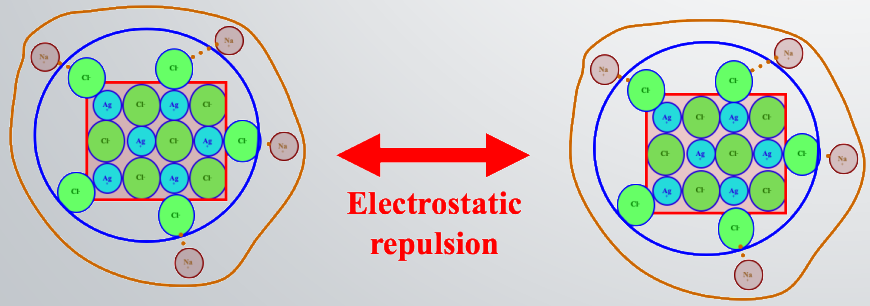
Excess Cl^-: $ClAg : Cl^- \cdots\cdots Na^+$

lattice 1st 2nd

Excess Ag^+: $AgCl : Ag^+ \cdots\cdots NO_3^-$

lattice 1st 2nd

2nd ionic layer can be tight or loose



Weak repulsion

Strong repulsion

2.6-4 Washing and filtering

Questions:

- Why do we wash the precipitate?

Remove droplets of liquid containing excess solute

- Which liquid can be used for washing?

Water or electrolyte?

What happens upon washing with pure water?

Peptization and losses upon filtration. **Why?**

Washing with pure water

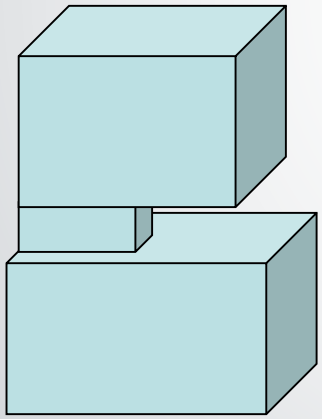
Before washing

Coagulated precipitate

2nd ionic layer is tight

$AgCl:Ag^{+} \cdot\cdot NO_3^{-}$
ClAg
$AgCl:Ag^{+} \cdot\cdot NO_3^{-}$

During washing:



$AgCl:Ag^{+} \cdots\cdots\cdots NO_3^{-}$
ClAg ↕ **Repulsion**
$AgCl:Ag^{+} \cdots\cdots\cdots NO_3^{-}$

The 2nd ionic layer is diluted. The charged solid particles repel one another and the precipitate breaks up---peptization.

Selecting the washing liquid

1. Electrolyte of an ion keep identical to the 2nd ionic layer (NO_3^-)
2. Use volatile electrolyte, so that it will evaporate during drying

Examples: HNO_3, NH_4NO_3 as the washing liquid for

$AgCl:Ag^+ \cdot\cdot NO_3^-$
ClAg
$AgCl:Ag^+ \cdot\cdot NO_3^-$

$$AgCl:H^+ \cdots NO_3^- \xrightarrow{Heat} AgCl\downarrow + HNO_3\uparrow$$

$$AgCl:NH_4^+ \cdots NO_3^- \xrightarrow{Heat} AgCl\downarrow + NH_3\uparrow + HNO_3\uparrow$$

What is preferred?

Washing with one large portion

or

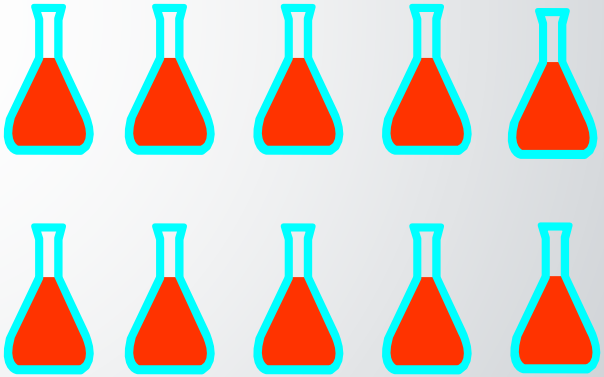
Washing with many small portions

A

1 Liter

or

B



10 × 0.1 Liter

Contaminants in the precipitate's liquid after *n* washing cycles

$$X_n = X_0 \left(\frac{u}{u+v} \right)^n$$

X_n = Contaminants in the precipitate's liquid after n washing cycles.

X_0 = Initial contaminants.

u = The volume of precipitate's liquid.

v = Volume used for each washing cycle.

n = Number of washing cycles.

Comparison

Contaminants in precipitate's liquid after one washing cycle with volume V

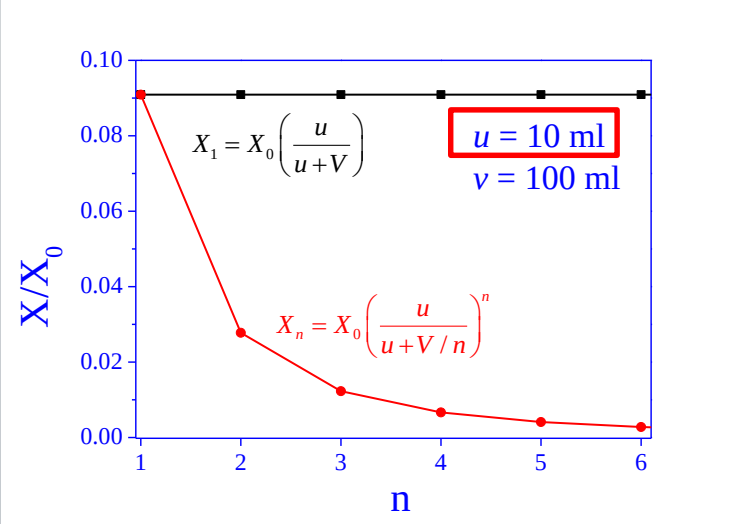
$$X_1 = X_0 \left(\frac{u}{u+V} \right)$$

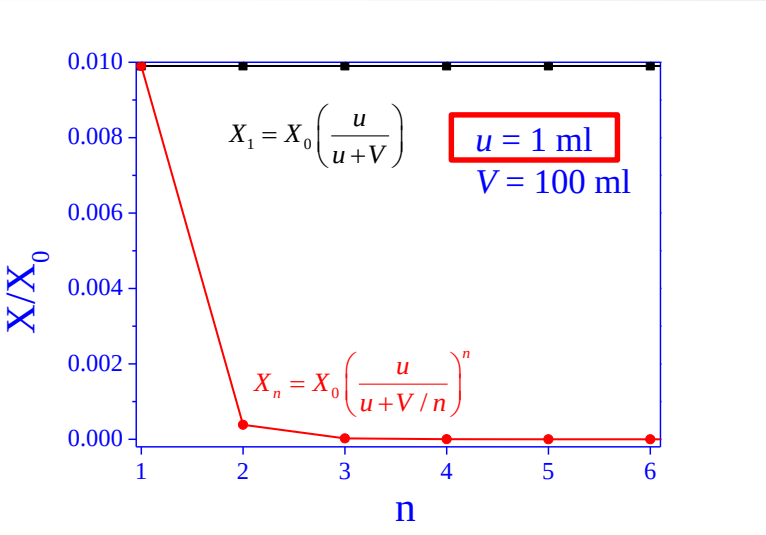
Contaminants in precipitate's liquid after n washing cycles with volume $v = V/n$

$$X_n = X_0 \left(\frac{u}{u+V/n} \right)^n$$

For $n = 1$ the expressions are identical.

For $n > 1$ always $X_n < X_1$



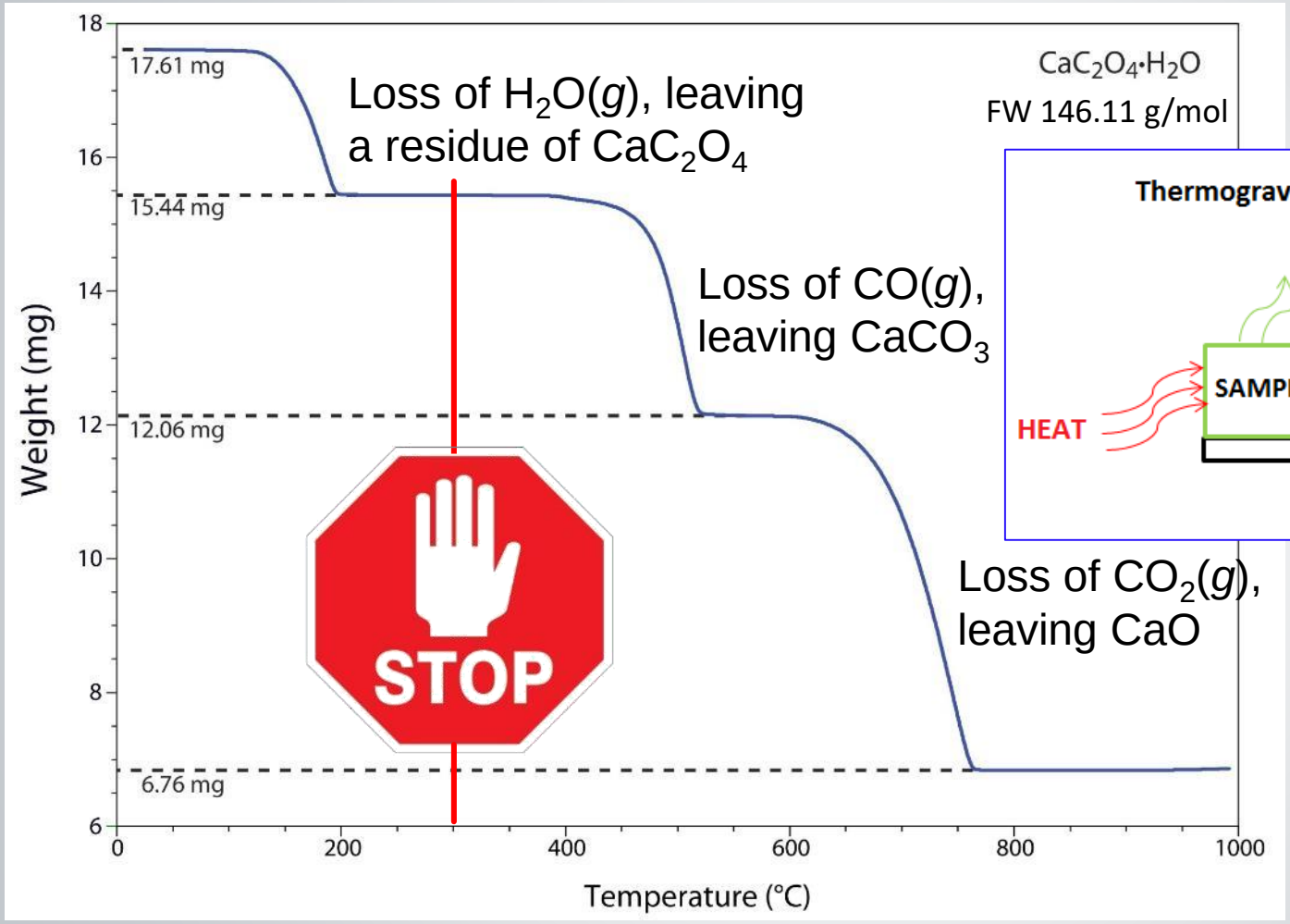


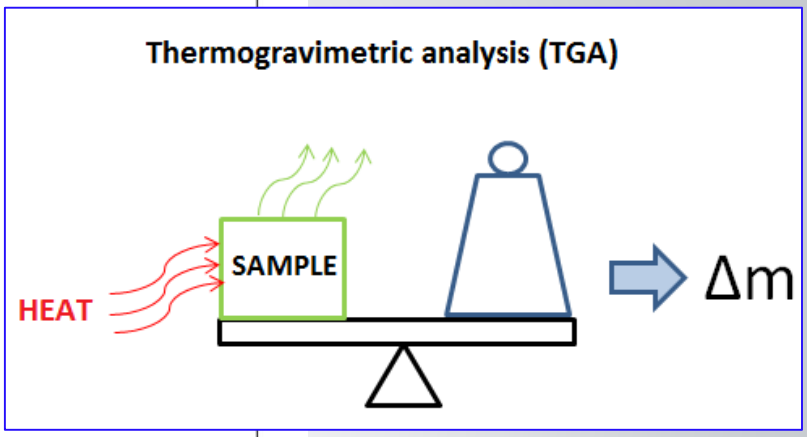
Heating the precipitate

Purposes

- Evaporation of water
- Evaporation of electrolyte in washing
- Moving to a component for weighing

Chose the heating temperature





Thermogram for $CaC_2O_4{\cdot}H_2O$ obtained by heating a sample from room temperature to 1000°C